SISTEMA DE METROLOGIA LEAN TECH – ARQUITETURA MODULAR DO NO PRINCIPAL (CARRIER BOARD)

Diagrama de Blocos Hierarquicos com Fluxo Sinalizacao Inputs/AFE -> MCU ESP32-S3 -> Outputs/LoRa

Power_Supply

VBAT_ADC

File: 01_Power_Supply.kicad_sch

Sensors_AFE

PWR_GATE_EN
AMP_ADC
TACHO
1W_DQ
I2S_SD
I2S_SCK
I2S_WS
I2C_SCL
I2C_SDA

File: 02_Sensors_AFE.kicad_sch

MCU_ESP32S3

VBAT_ADC
PWR_GATE_EN
AMP_ADC
TACHO
1W_DQ
I2S_SD
I2S_SCK
I2S_WS
I2C_SCL
I2C_SDA
LORA_M0
LORA_M1
LORA_TX
LORA_RX
LORA_AUX
GPIO_LED_GRN
GPIO_LED_ORG
GPIO_LED_BLU
GPIO_BUZZER

File: 03_MCU_ESP32S3.kicad_sch

Peripherals_LoRa

LORA_M0
LORA_M1
LORA_TX
LORA_RX
LORA_AUX
GPIO_LED_GRN
GPIO_LED_ORG
GPIO_LED_BLU
GPIO_BUZZER
I2C_SCL
I2C_SDA

File: 04_Peripherals_LoRa.kicad_sch

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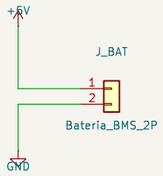
Sheet: /
File: amemiya_main_node.kicad_sch

Title: Amemiya Metrology – Main Node Carrier ESP32-S3 (Arquitetura Modula

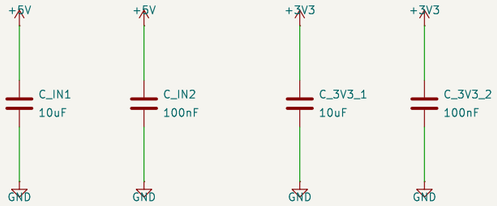
Size: A3	Date: 2026-10-16	**Rev: v36.2**
KiCad E.D.A. 9.0.7		Id: 1/5

ENTRADA DE ENERGIA 5V DO MODULO STEP-UP (BOOST)

Bateria 3.7V Li-Ion -> BMS TP4056 -> Chave ON/OFF -> Step-Up Boost 5V -> J_BAT



DESACOPLAMENTO DA ALIMENTACAO (+5V & +3V3)

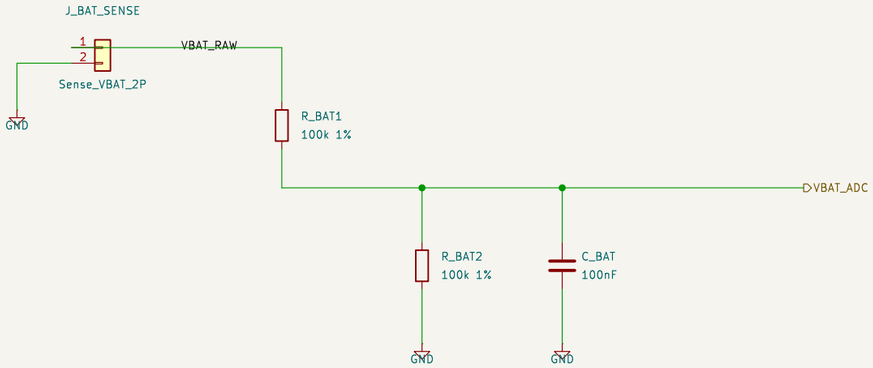


FONTES DE ALIMENTACAO DO SISTEMA (PWR_FLAG)



CIRCUITO DE MONITORAMENTO DE BATERIA (FUEL GAUGE DIVIDER)

Divisor 1:2 (100k / 100k) com filtro 100nF para leitura ADC de VBAT (0-4.2V -> 0-2.1V)



CADEIA DE ALIMENTACAO INDUSTRIAL EXTERNA (OFF-BOARD)

1. BATERIA: 2x Celulas 18650 Li-Ion (3.7V nom / 4.2V max / ~5200mAh em paralelo)

2. BMS/CARREGADOR: Modulo TP4056 com protecao contra sobrecarga e descarga profunda

3. CHAVE GERAL: Chave alavanca liga/desliga de painel (interrompe o polo positivo)

4. STEP-UP (BOOST): Modulo MT3608 ajustado para 5.00V DC regulado (eficiencia ~93%)

5. ENTRADA NA PLACA: Conector JST-XH 2P (J_BAT) -> Pino 22 do DevKit (5V IN)

6. SENSE DE BATERIA: Conector JST-XH 2P (J_BAT_SENSE) ligado antes do Step-Up

7. FORMULA FUEL GAUGE: V_ADC = V_BAT * (100k / (100k + 100k)) = V_BAT / 2 (Max 2.10V)
   -> Compativel com faixa linear do ADC1 (0 a 3.1V com atenuacao 11dB no ESP32-S3)

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Sheet: /Power_Supply/
File: 01_Power_Supply.kicad_sch

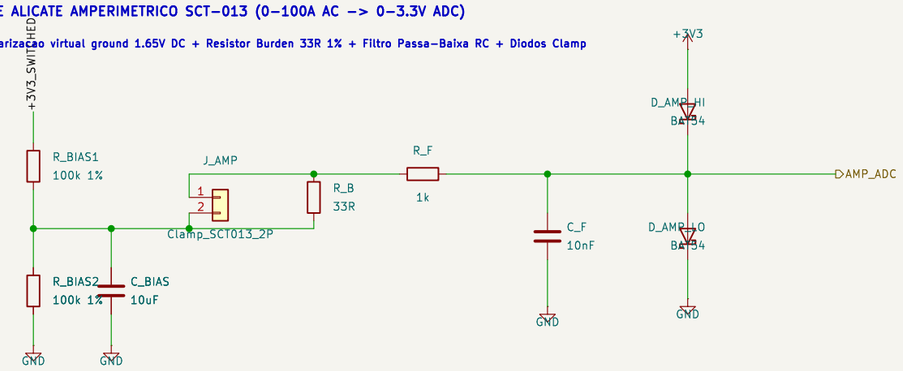
Title: 01 - Power Supply, Decoupling & Battery Fuel Gauge

Size: A3	Date: 2026-09-26	**Rev: v36.1**
KiCad E.D.A. 9.0.7		Id: 2/5

CONDICIONAMENTO DE SINAIS (AFE) E INTERFACES DE SENSORES INDUSTRIAIS

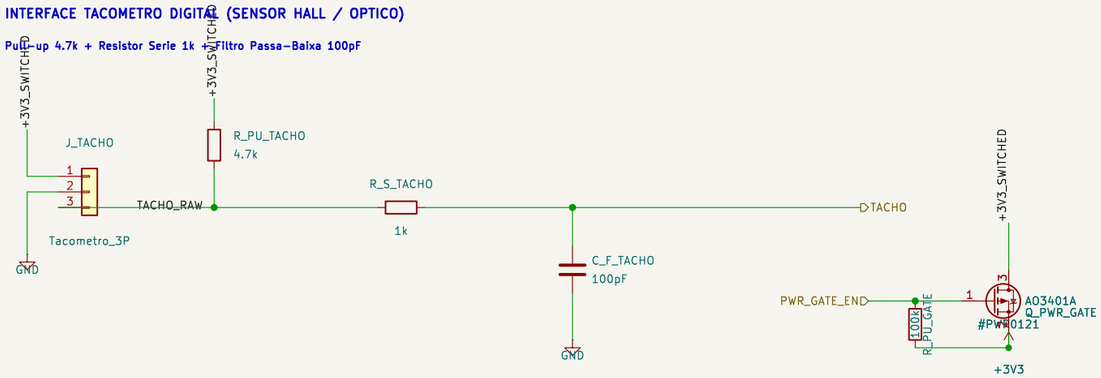
AFE ALICATE AMPERIMETRICO SCT-013 (0-100A AC -> 0-3.3V ADC)

Polarizacao virtual ground 1.65V DC + Resistor Burden 33R 1% + Filtro Passa-Baixa RC + Diodos Clamp



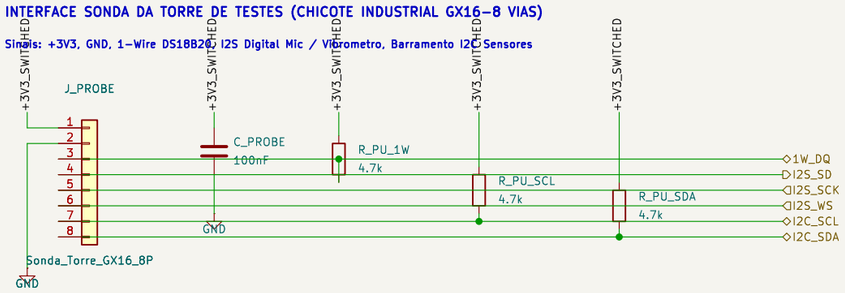
INTERFACE TACOMETRO DIGITAL (SENSOR HALL / OPTICO)

Pull-up 4.7k + Resistor Serie 1k + Filtro Passa-Baixa 100pF



INTERFACE SONDA DA TORRE DE TESTES (CHICOTE INDUSTRIAL GX16-8 VIAS)

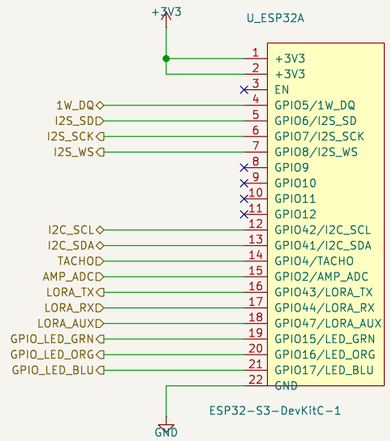
Sinais: +3V3, GND, 1-Wire DS18B20, I2S Digital Mic / Vibrometro, Barramento I2C Sensores

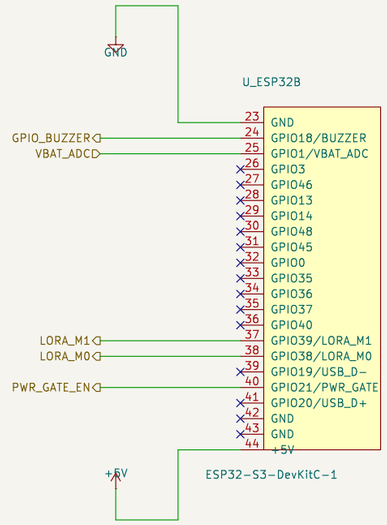


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Sheet: /Sensors_AFE/ File: 02_Sensors_AFE.kicad_sch		
Title: 02 – Analog Front-End (AFE) & Sensor Interfaces		
Size: A3	Date: 2026-09-26	**Rev: v36.1**
KiCad E.D.A. 9.0.7		Id: 3/5

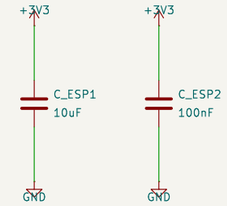
INTERFACES DE SOQUETE ESP32-S3 DEVKITC-1 (2x BARRA DE PINOS FEMEA 1x22, 2.54mm)

O DevKit conecta-se modularmente a baseboard, preservando USB-C, LDO e botoes onboard.





DESACOPLAMENTO LOCAL PROXIMO AOS SOQUETES



MAPEAMENTO DE PINOS E PERIFERICOS DO ESP32-S3 (DEVKITC-1)

GPIO 01 (ADC1_CH0): Leitura de Tensao da Bateria (VBAT_ADC via divisor 1:2)

GPIO 02 (ADC1_CH1): Sinal AC do Alicate Amperimetrico (AMP_ADC com bias 1.65V)

GPIO 04: Entrada de Pulsos do Tacometro Optico/Hall (TACHO com filtro RC)

GPIO 05: Barramento 1-Wire Termometro DS18B20 da Torre (1W_DQ + pull-up 4.7k)

GPIO 06 / 07 / 08: Barramento I2S do Microfone/Vibrometro (I2S_SD, I2S_SCK, I2S_WS)

GPIO 15 / 16 / 17: LEDs de Painel (Verde=RUN, Laranja=STANDBY, Azul=TELEMETRIA)

GPIO 18: Buzzer Ativo Industrial para Alarmes e Beeps de Operacao (GPIO_BUZZER)

GPIO 41 / 42: Barramento I2C compartilhado (I2C_SDA, I2C_SCL para MPU6050 & OLED)

GPIO 43 / 44 / 47: UART Transceptor LoRa E32-TTL-100 (LORA_RX, LORA_TX, LORA_AUX)

ALIMENTACAO: 5V IN no Pino 22 (J3) -> LDO onboard do DevKit produz +3.3V nos Pinos 1 e 2 (J1)

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Sheet: /MCU_ESP32S3/

File: 03_MCU_ESP32S3.kicad_sch

Title: 03 - ESP32-S3 DevKit 44-Pin Dual Socket Interface

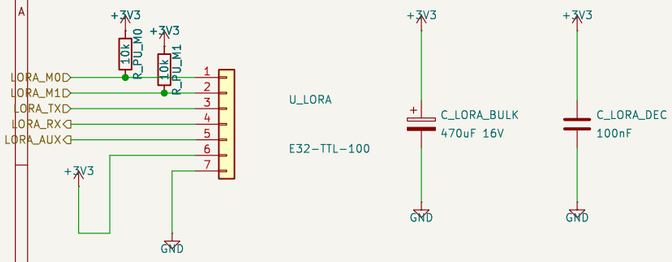
Size: A3 | Date: 2026-09-26 | **Rev: v36.1**

KiCad E.D.A. 9.0.7 | Id: 4/5

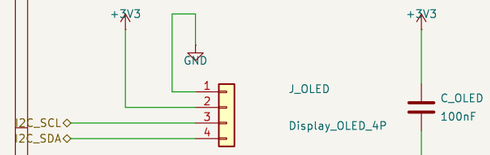
COMUNICACAO RF LORA EBYTE E32-TTL-100 / ATUADORES DE INTERFACE HUMANO-MAQUINA

TRANSCEPTOR LORA EBYTE E32-TTL-100 (433MHz / 915MHz)

Capacitor bulk 470uF Low-ESR essencial para absorver surtos de transmissao RF sem brownout

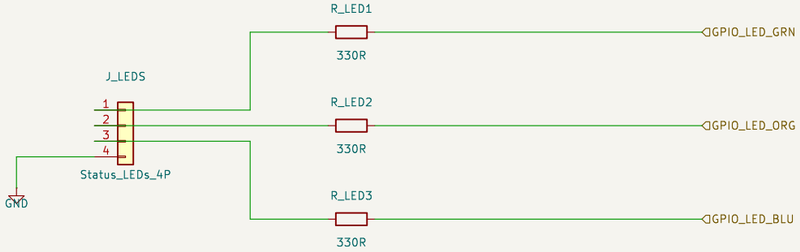


DISPLAY OLED GRAFICO 0.96 POL SSD1306 I2C





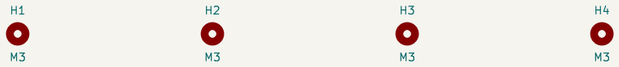
LEDS DE STATUS DO PAINEL (VERDE, LARANJA, AZUL)



BUZZER SONORO ATIVO DE ALERTA INDUSTRIAL



FUROS DE FIXACAO MECANICA (M3) NOS 4 CANTOS DA PLACA (90x65mm)



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Sheet: /Peripherals_LoRa/
File: 04_Peripherals_LoRa.kicad_sch

Title: 04 – LoRa E32 RF, OLED Display, Status LEDs & Buzzer

Size: A3	Date: 2026-09-26	**Rev: v36.1**
KiCad E.D.A. 9.0.7		Id: 5/5